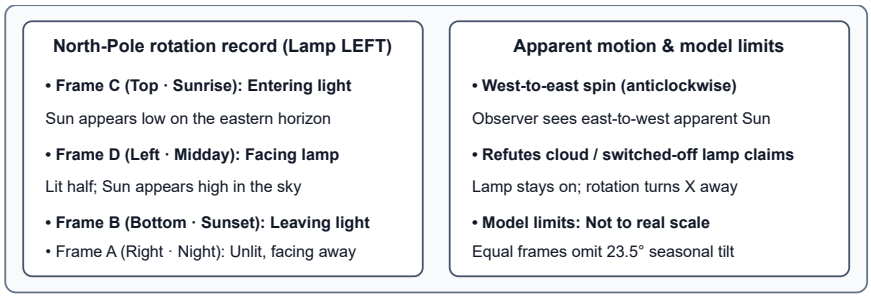
Reference sheets

Reference A · Four-frame North-Pole rotation record (Fixed lamp on LEFT)

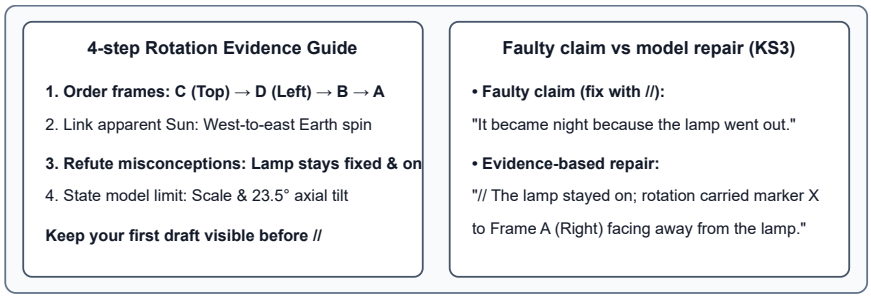


North-Pole rotation record. It links Frames C, D, B, and A to apparent Sun motion and limits.

Reference only. Do not write on this sheet. Use these supplied observations to support your rotation explanation.

Ordered frame & position.	Illumination state of marker X.	Apparent Sun position from marker X.	Rotation clue & model note.
1. Frame C · Top (Sunrise).	Entering the lit left half from shadow.	Sun appears low on the eastern horizon.	Start of cycle (0°); Earth spins anticlockwise (west to east).
2. Frame D · Left (Middle of day).	Facing the fixed lamp directly in the lit half.	Sun appears high across the sky.	After 1st quarter turn (90°); lamp stays fixed and on.
3. Frame B · Bottom (Sunset).	Leaving the lit left half into shadow.	Sun appears low on the western horizon.	After 2nd quarter turn (180°); observer turns away from lamp.
4. Frame A · Right (Night).	Facing away from the lamp in the unlit right half.	Sun is below the local horizon (darkness).	After 3rd quarter turn (270°); 4th turn returns X to Top (360°).

Reference B · Refuting misconceptions, apparent motion, and model limits



Four-step Rotation Evidence guide. It shows frame ordering, apparent motion, refuting cloud claims, and limits.

Evidence step.	Sentence stem for your book.	Key Stage 3 check.
1. Order the 24-hour cycle.	From sunrise, marker X moves through C (Top), D (Left), B (Bottom), and A (Right).	Did you track the same marker back to its starting point?
2. Explain apparent Sun motion.	Earth rotates west to east, so the Sun appears to move east to west.	Did you distinguish observer viewpoint from real solar motion?
3. Refute 'clouds/lamp off' claims.	No clouds are present and the lamp stays on. Rotation turns X into shadow.	Did you use Frame D and Frame A to prove orientation changed?
4. State one honest model limit.	// Limit: size and distance are not to scale. Equal frames omit 23.5° tilt.	Did you state what equal quarter turns cannot show?

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Box 1 of the arrival drawing. Why is the daily day-night cycle caused by rotation rather than Earth's yearly orbit?

Help. Compare the 24-hour spin with the yearly orbital path.

2. Look at Box 2 (North-Pole view, lamp on left). Marker X moves anticlockwise from Bottom to Right. Which transition occurs?

Help. Follow the anticlockwise arrow from Bottom to Right.

3. Why is apparent Sun motion across the sky different from the Sun orbiting Earth?

Help. Distinguish an observer's viewpoint from physical motion.

4. Look at Box 3. Do four equal model frames prove that daylight lasts 12 hours everywhere on Earth?

Help. Consider how tilt and latitude change real daylight length.

You do tasks

1. Use Reference A. Construct a four-frame North-Pole evidence account for marker X. Link anticlockwise rotation to apparent east-to-west solar motion. **Staff I ☐ S ☐ C ☐**

Help. Name each frame, its illumination state, and the rotation mechanism.

2. Use Reference B. Distinguish observation from inference to refute 'Clouds cause night.' Explain why a moving lamp could mimic the pattern but represent false physics.

Staff I ☐ S ☐ C ☐

Help. Separate what is observed in Frames D and A from the physical cause.

3. Evaluate one strength and one limit of the four-frame model. Add a two-part // revision after feedback. **Staff I ☐ S ☐ C ☐**

Help. Link the model limit to latitude or seasonal axial tilt.

Optional participation-loop prompt. Share your four-frame evidence and limit with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Use Frames D and A to refute 'Clouds cause night.' Separate observation from inference.

2. A moving torch around a still globe gives a similar light pattern. Why is that physical cause wrong?

3. What real daylight question cannot be answered by four equal 90-degree North-Pole frames alone?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt